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TITLE: EcoBeacon – Smart Carbon Footprint Tracking and Reduction System

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ABSTRACT

EcoBeacon AI is an AI-powered web application designed to help users track, analyze, and reduce their carbon footprint generated from daily activities such as transportation, electricity usage, food consumption, and shopping. The system is being developed using the MERN stack and currently includes implemented modules like user authentication, activity logging, database integration, and a carbon emission calculation engine based on standardized emission factors. It generates an eco score and provides basic AI-driven suggestions for sustainable practices. The frontend dashboard for data visualization is under development, with future enhancements including advanced AI features, image-based tracking, and gamification. The platform aims to promote environmental awareness and encourage eco-friendly lifestyle choices through an interactive and user-friendly approach.

1.1 Problem Background

Climate change continues to intensify due to rising greenhouse gas emissions from everyday human activities such as transportation, electricity consumption, dietary habits, and shopping behaviour. Although awareness regarding sustainability is increasing, individuals lack practical tools to measure and manage their personal carbon footprint.

Existing solutions suffer from:

- Lack of **user-friendly interfaces**
- Minimal **personalization**
- Absence of **real-time tracking**
- Limited **engagement features**

This creates a gap between awareness and actionable sustainability practices.

1.2 Objective of the Project

The objective of **EcoBeacon AI** is to design and implement a **fully functional smart web application** that:

- Tracks user activities in real-time
- Calculates carbon emissions using standardized datasets
- Provides **AI-driven personalized suggestions**
- Encourages sustainable behavior using **gamification and dashboards**

1.3 Progress Made Since First Evaluation

Module	Status	Description
User Authentication	Completed	Login & Signup using JWT authentication
Activity Logging Module	In Progress	Users can log travel, food, electricity, shopping
Carbon Calculation Engine	In Progress	Backend logic implemented using emission factors
Dashboard UI	In Progress	Graphs and charts partially integrated

AI Recommendation System	In Progress	Basic rule-based suggestions implemented
Database Design	Completed	MongoDB schema created and integrated
Frontend UI	In Progress	Built using React + Tailwind CSS

1.4 System Architecture

Layer	Technology Used	Description
Frontend	React.js, Tailwind CSS	User interface and dashboards
Backend	Node.js, Express.js	API handling and business logic
Database	MongoDB	Stores user data and activity logs
AI Module	Python / JS Logic	Recommendation engine
APIs	REST APIs	Communication between frontend & backend

1.5 Proposed System Workflow

Step Process

- 1 User logs into the system
- 2 Inputs daily activities
- 3 Data stored in MongoDB
- 4 Carbon emission calculated using backend logic
- 5 Eco score generated
- 6 AI module suggests improvements
- 7 Dashboard displays insights

1.6 Key Features (Enhanced Version)

Feature	Description
Carbon Calculator	Calculates emissions based on activities
Eco Score	Score to track sustainability level
Dashboard	Visual graphs for emission trends
AI Suggestions	Personalized tips to reduce footprint
Activity Tracking	Daily logging system
Image-based Estimation	(In Progress) Detect activities via images
Gamification	Rewards, badges (Planned)

1.7 Sample Carbon Emission Data Table

Activity Type	Unit	Emission Factor (kg CO ₂)
Car Travel	per km	0.12
Electricity	per kWh	0.85
Meat Consumption	per meal	2.5
Public Transport	per km	0.05

1.8 Database Schema Overview

Collection	Fields
Users	userId, name, email, password
Activities	userId, type, value, date
Emissions	userId, totalEmission, ecoScore
Suggestions	userId, recommendation

1.9 Expected Outcome and Benefits

- Provides **real-time carbon tracking**
- Encourages **eco-friendly decision-making**
- Increases **user awareness through visualization**
- Promotes **behavioral change using gamification**
- Helps organizations monitor sustainability efforts

1.10 Original Contribution Statement

EcoBeacon AI goes beyond traditional carbon trackers by integrating:

- **Real-time activity-based tracking**
- **AI-driven personalized recommendations**
- **Interactive dashboards**
- **Planned image-based estimation and gamification**

This makes the system more **practical, engaging, and scalable** for real-world usage.